

- A (110)₂ = Load AC from Memory Address
 B (1011)₂ = Store AC to Memory Address
 C (1100)₂ = Add AC from Memory Address

Each instruction has 4-bit opcode and 12-bit address, which identifies a particular I/O device or memory address. Show the program execution using the following format for the following program. Program is stored in memory starting at address 300. All address locations are in hexadecimal.

1. Load AC from memory address 939 - A939
2. Add contents of memory address 940. - C940
3. Store AC to memory address 941 - B941

Assume that memory location 939 contains a value of 000B and memory location 940 contains a value of 0003. Use hexadecimal numbers for instructions and data. Register AC and IR are 16-bit wide while register PC is 12-bit wide. Fill the memory locations and CPU registers with appropriate values in hexadecimal for all six steps given below.

Memory CPU Registers

300	A939	PC	300
301	C940	AC	
302	B941	IR	A939
939	000B		
940	0003		
941			

STEP 1

Fetch

Memory CPU Registers

300	A939	PC	301
301	C940	AC	000B
302	B941	IR	A939
939	000B		
940	0003		
941			

STEP 2

Exe

Memory CPU Registers

300	A939	PC	301
301	C940	AC	000B
302	B941	IR	C940

939	000B
940	0003
941	

STEP 3

Fetch

Memory CPU Registers

300	A939	PC	302
301	C940	AC	000E
302	B941	IR	C940

939	000B
940	0003
941	

000B
+ 0003

STEP 4

Exe

Memory CPU Registers

300	A939	PC	302
301	C940	AC	000E
302	B941	IR	B941

939	000B
940	0003
941	

STEP 5

Fetch

Memory CPU Registers

300	A939	PC	303
301	C940	AC	000E
302	B941	IR	B941

939	000B
940	0003
941	000E

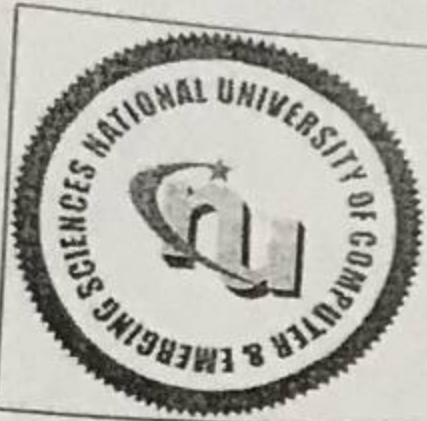
STEP 6

Exe

0000--0
 0001--1
 0010--2
 0011--3
 0100--4
 0101--5
 0110--6
 0111--7
 1000--8
 1001--9
 1010--A
 1011--B
 1100--C
 1101--D
 1110--E
 1111--F

RW

$$\begin{array}{r} 1011 \\ + 0011 \\ \hline 1110 = E \end{array}$$



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Instruction/Notes: Solve All Questions. Attempt all parts of a question together.

Q.1 Consider a hypothetical 32-bit microprocessor having 32-bit instructions composed of two fields; the first byte contains the opcode and the remainder the immediate operand or an operand address.

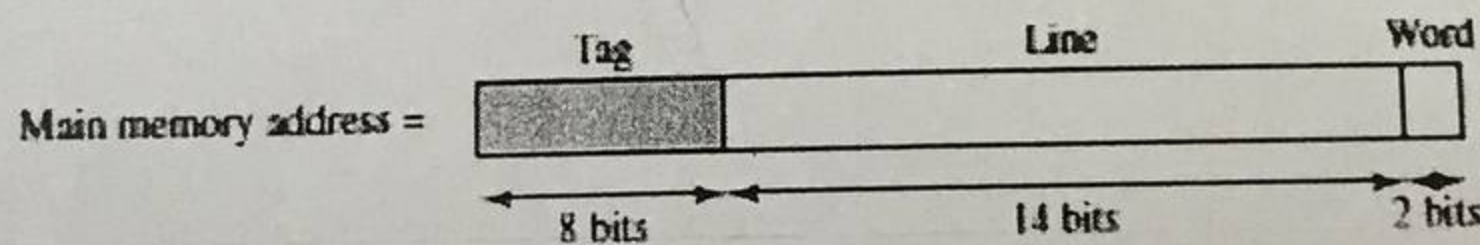
PC = 12 bits IR = 16 bits

- What is the maximum directly addressable memory capacity (in bytes)? 16
- How many bits are needed for the program counter and the instruction register?
- What architectural features will allow this microprocessor to access a separate "I/O space"?
- If an input and an output instruction can specify an 16-bit I/O port number, how many 16-bit I/O ports can the microprocessor support? [CLO 1][10]

Q.2 a. Suppose an 8-bit data word stored in memory is 00110001. Using the Hamming algorithm with even parity, determine what check bits would be stored in memory with the data word. Give the complete codeword in hexadecimal. Show how you got your answer. [CLO 2][5]

b. With 1-Mbyte chip, where the word size is 1 byte. Give a high-level chip diagram of how to construct an 8-Mbyte computer memory using eight 1-Mbyte chips. Be sure to show the address lines in your diagram and what the address lines are used for. Specify memory address range for each chip (memory map). [CLO 2][5]

Q.3 For a direct mapped cache memory system, main memory address is defined as: [CLO 2][10]



Determine the following values:

- number of addressable units. 2^{24}
- block size. 4 bits
- number of blocks in main memory. 2^{22}
- number of lines in cache. 2^{14}

$$iv - m = 2^n$$

$\begin{array}{r} 111 \\ 100 \\ \hline -8+1 \end{array}$

National University of Computer and Emerging Sciences – Lahore Campus
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Q.1 For the following 8-bit binary numbers perform Arithmetic Shift Right and Arithmetic Shift Left. Interpret the results in decimal. [CLO 4] [10]

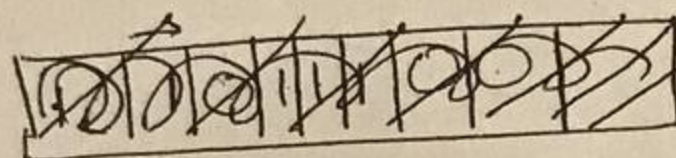
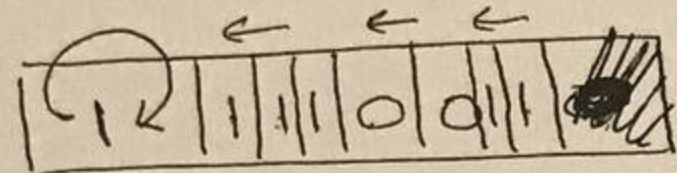
a) 11110011

b) 00001101

a) 11110011

ASL :

ASR :



sign bit preserved

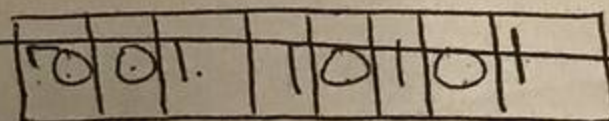
$$= \boxed{1 \mid 1 \mid 1 \mid 0 \mid 0 \mid 1 \mid 1 \mid 0} = (-154)_{10}$$

$$= (10111001)_2 = (-199)_{10}$$

(b) 00001101

ASL :

ASR :

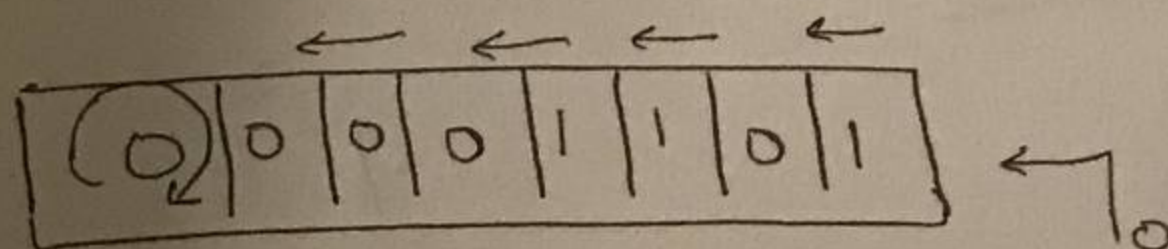


00001101

Part (b)

00001101

ASL:



$$= (00011010)_2 = (26)_{10}$$